Krishnaa **Vadivel** | Embedded Systems Engineer | Firmware | FPGA | Robotics

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Profile

Embedded systems engineer with experience spanning firmware development, FPGA logic, robotics, instrumentation, and field validation. Background includes low-level development for sensing and logging platforms, board bring-up and debugging, and companion software for analysis and visualization around embedded hardware.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Developed embedded functionality for geological well tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class hardware platforms.
- Implemented sensor-logging and control-side logic and firmware for field-deployed systems operating in instrumentation-heavy environments.
- Performed oscilloscope-based debugging, bench validation, and field testing to verify hardware-software integration and system behavior.
- Supported selective PCB updates in Altium and related board-level engineering tasks as system requirements evolved.
- Built companion host-side engineering tools using CUDA for waveform processing, visualization, and diagnostic workflows around the embedded stack.
- Contributed additional internal software using C#, ASP-style services, and Microsoft SQL-backed data workflows.

Yale University
Embedded Systems Teaching Assistant / PhD Researcher *2021–present*

- Supported embedded systems laboratories and student projects involving firmware, hardware interfaces, debugging practices, and instrumentation workflows.
- Continued broader engineering work in scientific computing and simulation while maintaining strong embedded-systems fluency.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on excited-state materials physics.

Selected Projects

EMG-Driven Robotic Hand

- Built an embedded robotic-hand project in undergraduate Cornell ECE work, integrating surface EMG sensing, analog filtering, PIC32 control logic, and servo actuation; related work was published in Circuit Cellar.

Maze Mapping Robot with Wireless Control

- Worked on a robotics project combining navigation logic, remote-station control, and wireless coordination in a course setting.

Single-Cycle MIPS Processor

- Implemented a small Verilog CPU and self-checking test bench to demonstrate digital design, datapath organization, and reproducible verification.

Matrix-Multiply Accelerator

- Built a compact Verilog accelerator block with clear start/done control behavior and a simulation-backed test flow.

Host-Side Diagnostic and Visualization Tooling

- Developed CUDA-based utilities for analyzing and visualizing waveform data associated with embedded sensing and logging systems.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Ongoing doctoral work; also served as an embedded systems teaching assistant.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*
Included advanced engineering coursework and embedded-adjacent systems work.

Cornell University
BS, Electrical and Computer Engineering *2013–2017*
Included embedded systems, robotics, and hardware-software integration foundations.

Selected Coursework

Embedded: Advanced Embedded Systems; laboratory-oriented firmware and systems work
Signals and Systems: Mathematics of Signals and Systems
Hardware: VLSI Design; Analog VLSI Design; digital and mixed engineering foundations
Supporting Theory: Semiconductor Physics; controls/robotics-adjacent systems exposure in ECE training

Technical Skills

Languages: C, C++, Python, Bash, JavaScript
Hardware: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, sensor logging, embedded control
Interfaces: SPI, I2C, UART, hardware-software integration workflows
Validation: Oscilloscopes, bench instrumentation, field testing, board bring-up, Altium update support
Software: Linux, CUDA, C#, ASP.NET, Microsoft SQL Server